

Cover Letter — Nature Communications

Asymmetric Structural Transfer Between Natural Language and Biological Foundation Models

Dear Editors,

We submit our manuscript, “**Asymmetric Structural Transfer Between Natural Language and Biological Foundation Models**,” for consideration as an Article in *Nature Communications*.

The question. Cross-domain transfer—reusing representations learned in one domain to solve tasks in another—is among the most consequential yet least understood properties of foundation models. A striking recent instance has been reported between human language and the “language of life”: models trained only on natural-language structure can, without any biological pre-training, discriminate homologous biological sequences. This has been read as evidence that abstract linguistic structure is a universal prior for biology. We ask a more basic question that this literature has left unexamined: **is such structural transfer symmetric?** If two domains truly share a structural substrate, transfer should run in both directions. We show it does not.

What we find. Through a systematic bidirectional study—fine-tuning across 100 seeds, iso-token continued pre-training with shuffled and random controls, model scaling, two independent biological foundation-model families (ESM-2 and ProtBERT), and adversarial synthetic structure tasks—we establish that structural transfer is **bidirectional but strongly and universally asymmetric**. Language→biology transfer is robust and strengthens with scale; biology→language transfer is weak, never exceeds matched-token controls, and *decays toward chance* as protein models grow. In an architecture-matched analysis on models with fully **known** training data, a language model retains roughly four times more competence when moved to biology (off-domain drop 0.08) than a biological model retains when moved to language (drop 0.36). We further show the forward effect is genuinely structural, not paraphrase-specific: three distinct language-structure tasks—including a purely syntactic bracket-matching task with no semantics—all transfer to protein homology.

Why it matters. The contribution is conceptual, not a benchmark result. First, it reframes a much-discussed phenomenon: language→biology transfer is not evidence of a *symmetric* universal grammar but of an *asymmetric* one. Second, it overturns a common intuition about scale—more parameters do not universally improve cross-domain transfer; here scaling *deepens domain specialization* and drives the two directions apart. Third, and most broadly, it establishes a principle likely to extend beyond this domain pair: **structural similarity between domains does not imply transferable capability**. Two domains can share statistical and structural regularities, and a model can be highly competent in each, without transfer between them being symmetric or even present.

Rigor and scope. Because our central comparisons use small models whose training corpora are fully documented (GPT-2, BERT, ESM-2, ProtBERT), the effect cannot be attributed to pre-training contamination—the confound that clouds studies relying on proprietary large models, which we report only as an exploratory upper bound. Mechanistic analyses (representational-alignment, learning-dynamics, and attention-circuit tests) localize the asymmetry to *learnability* rather than static representation. We believe the breadth of the question—spanning machine learning, computational biology, and the foundations of representation learning—suits the cross-disciplinary readership of *Nature Communications*.

The manuscript has not been published elsewhere and is not under consideration by another journal. All data, code, and the from-scratch control model are publicly released. The author declares no competing interests. We thank you for your consideration.

Sincerely,

Liang Wang

School of Artificial Intelligence and Automation,

Huazhong University of Science and Technology

wangliang.f@gmail.com